Introduction to Ambient Intelligence

Ambient Intelligence and Domotics

Master Degree: Artificial Intelligence for Science and Technology

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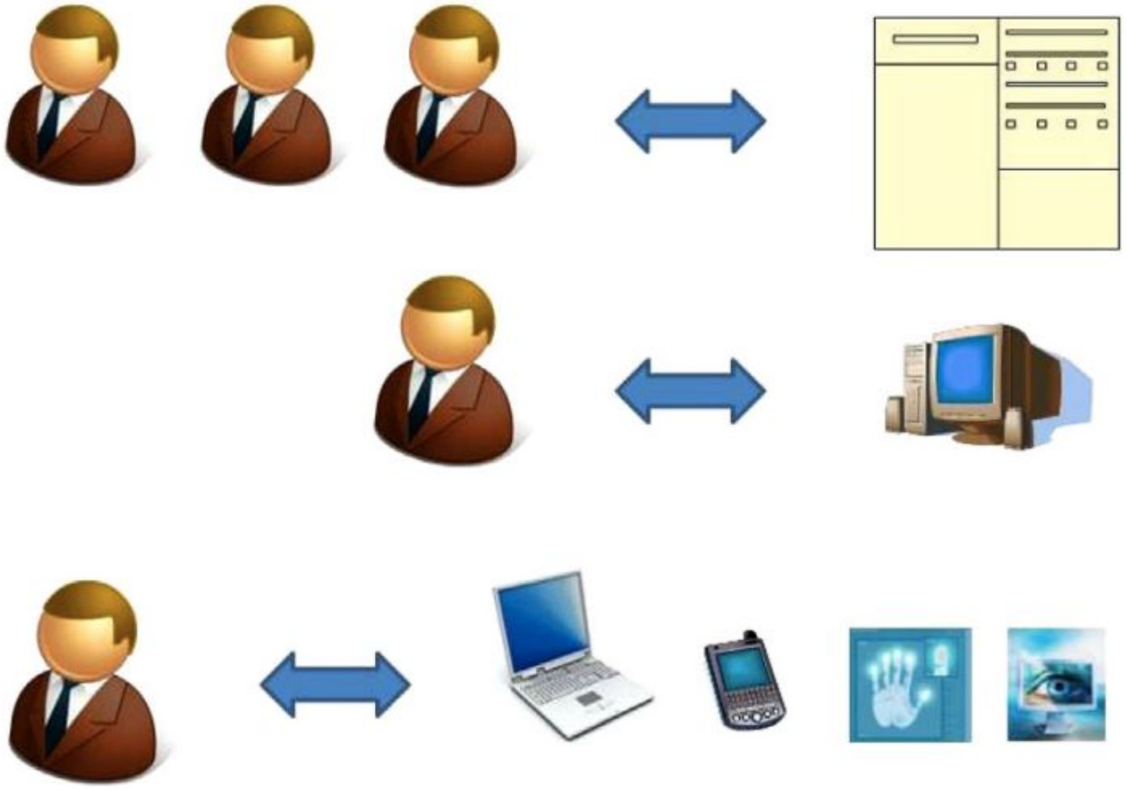
What is Ambient Intelligence?

People-computing ratio shift

- Technological evolution in the last decades led to significant shifts in the people-computing ratio
- Nowadays, each person has multiple personal computing devices besides the PC
  - smartphone
  - wearables
  - smart home appliances
  - domotic devices
  - …



Cook, D. J., Augusto, J. C., & Jakkula, V. R. (2009). Ambient intelligence: Technologies, applications, and opportunities. Pervasive and mobile computing, 5(4), 277-298.

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Ambient Intelligence

- ***Ambient Intelligence*** is a multidisciplinary area with the objective of making everyday environments **intelligent** and **adaptive**
  - A network of ***"hidden interfaces"*** that understands our behavior and adapts the environment based on our needs
  - The environment is indeed **aware** of the current ***context***, and ready to **act**

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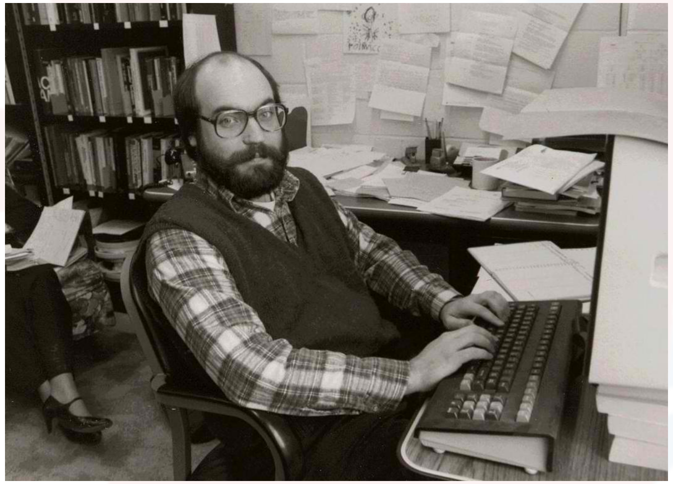
The vision

“The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it”

Mark Weiser, 1988

The Computer for the 21st Century



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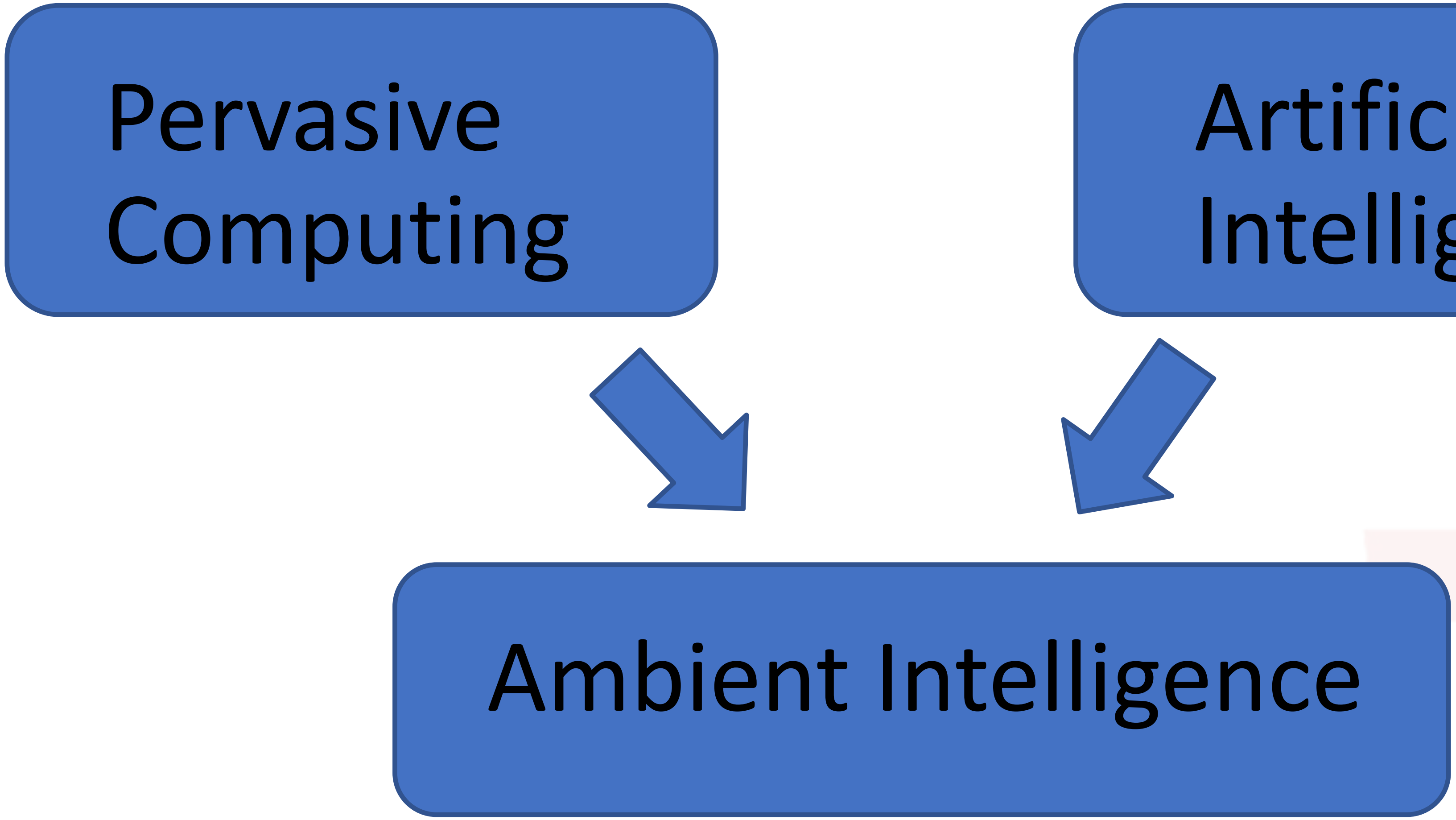
Pervasive Computing
Artificial Intelligence
Ambient Intelligence

AmI is based on Distributed Pervasive Systems

A distributed system is:

A collection of connected independent computing nodes that appears to its users as a single coherent system

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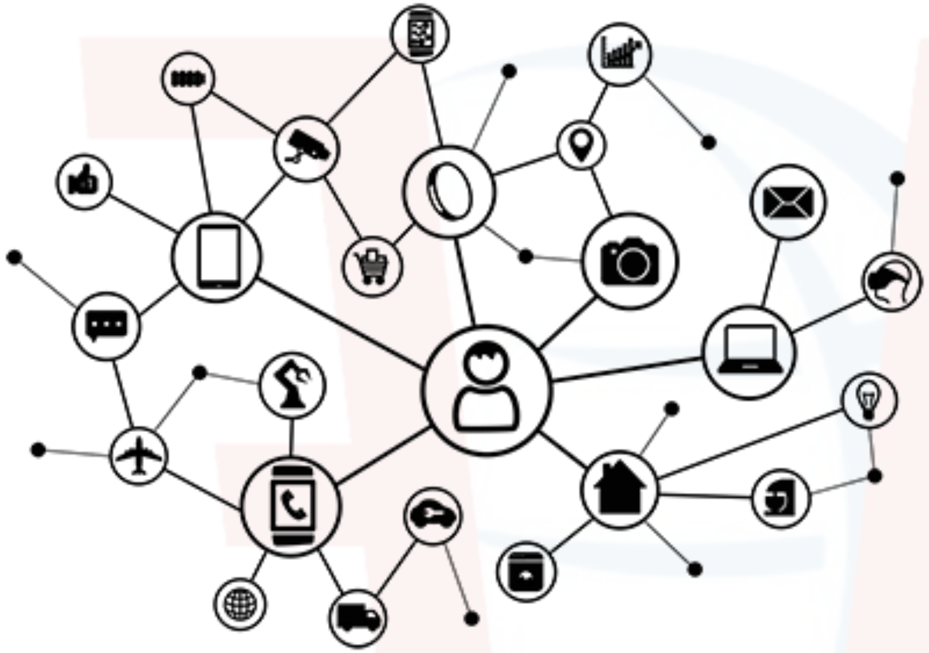
AmI is based on Distributed Pervasive Systems

A **Pervasive System** is a distributed system with the following main features:

- ***it includes unconventional nodes***
  - (possibly mobile) objects with ***computing*** and ***communication*** capabilities (smartphones, smart appliances, smart meters, sensors, ...)
- **it is adaptive**
  - the system considers the current context and adapts the behavior for optimizing the system goal



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Pervasive systems are ... volatile systems

Pervasive systems exhibit high volatility:

- failures of devices and communication systems
- changes in the characteristics of communication such as bandwidth
- creation and destruction of associations between software components resident on the devices

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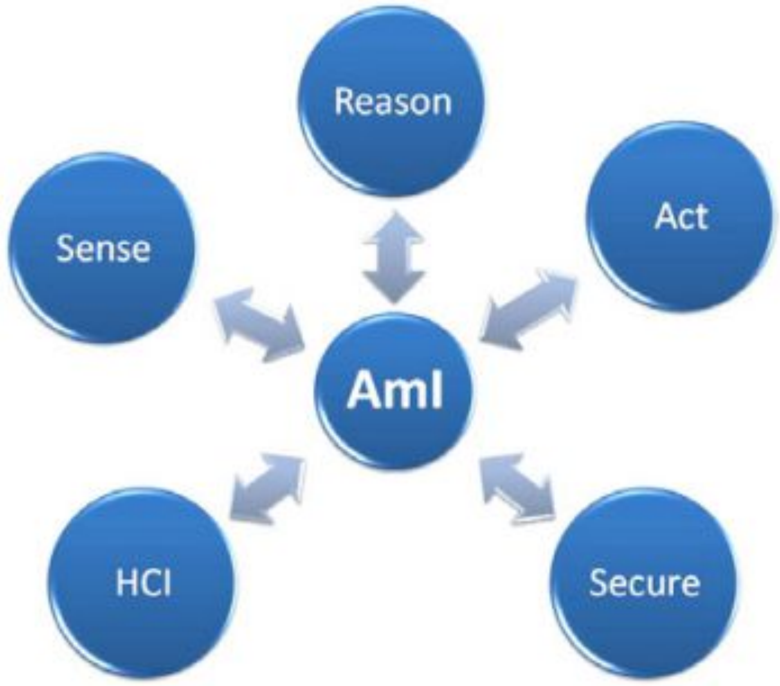
Ambient Intelligence (AmI) components

- **Sensing:**
  - physical components that capture information about the environment and the users living in it
- **Acting:**
  - physical components that act on environments (e.g., lights, thermostat, robots)
- **Reasoning:**
  - Modeling and recognizing the user’s behavior in the environment to connect sensing and acting
- **Human-Computer-Interaction (HCI):**
  - intelligent methods to simplify the interaction of the subject with the environment based on the sensed information (e.g., gestures, speech, emotions)
- **Security&privacy:**
  - sensed data about users may be sensitive and smart devices may be subject to attacks



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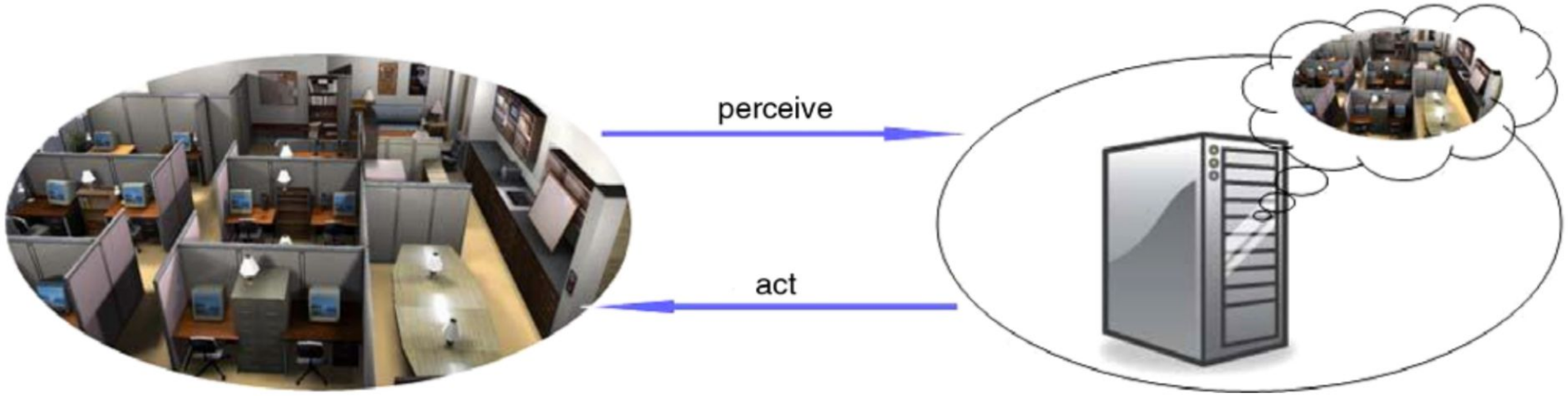
AmI and Artificial Intelligence

- ***<u>Understanding current context</u>***
- Predicting user needs
- Naturally interacting with users (NLP, gesture, ...)



Ambient Intelligence is based on IoT

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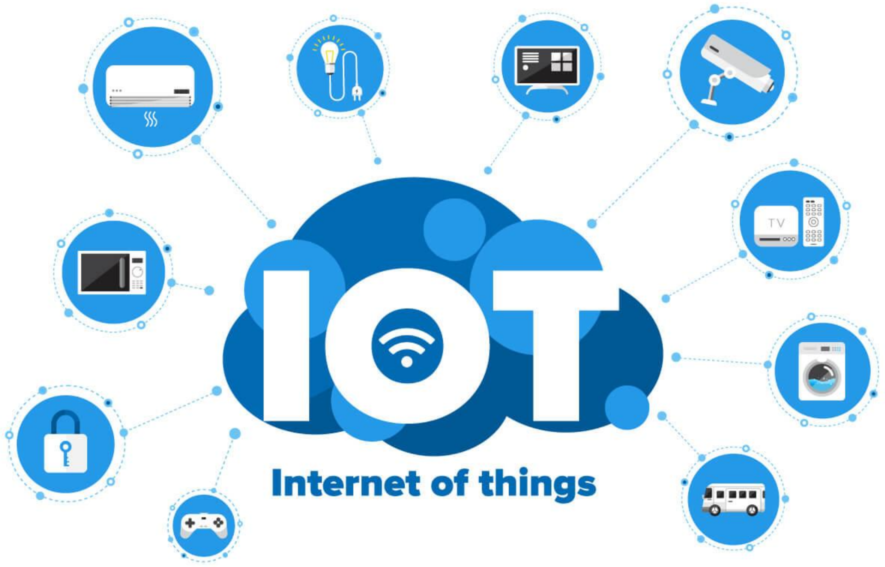
Internet of Things (IoT)

A set of physical “smart” objects (*things*) equipped with sensors, software, and other technologies, collecting data from the environment and exchanging it over a computer network to achieve a specific goal.



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Why "smart" objects?

- Connected to the Internet
  - remote access, invocation of services, cooperation, ...
- Running algorithms (locally or remotely) to analyse data and understand "context" by exploiting AI techniques.
- Offering *personalized context-aware* services

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"Smart" Appliances

Smart … everything

SmartWatch, SmartScale, SmartLocks, SmartSleep …

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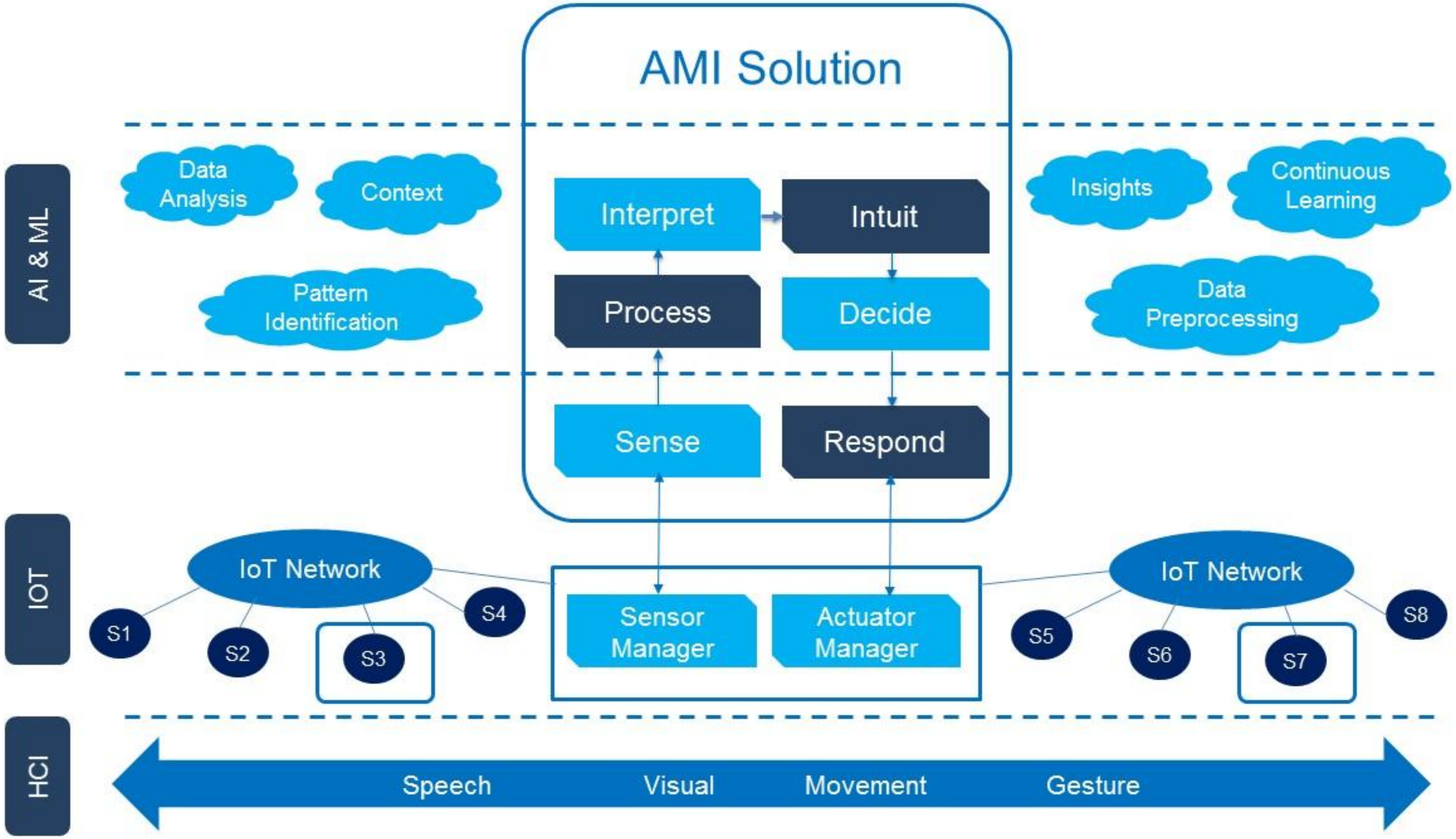
Role of IoT in Ambient Intelligence



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IoT + Home = Smart-Homes (Domotics)

- Domotics is the use of IoT devices (sensors and actuators) in the home environment, to transform it in a smart-home
- Usual application for common users: automations
- Example:
  - Status of luminance in the living room: low
  - Events happen:
    - Motion detected in living room
  - Automation (rule that changes device states based on events)
    - When motion detected in living room and luminance is low, turn on the light
  - Effect:
    - Light is turned on

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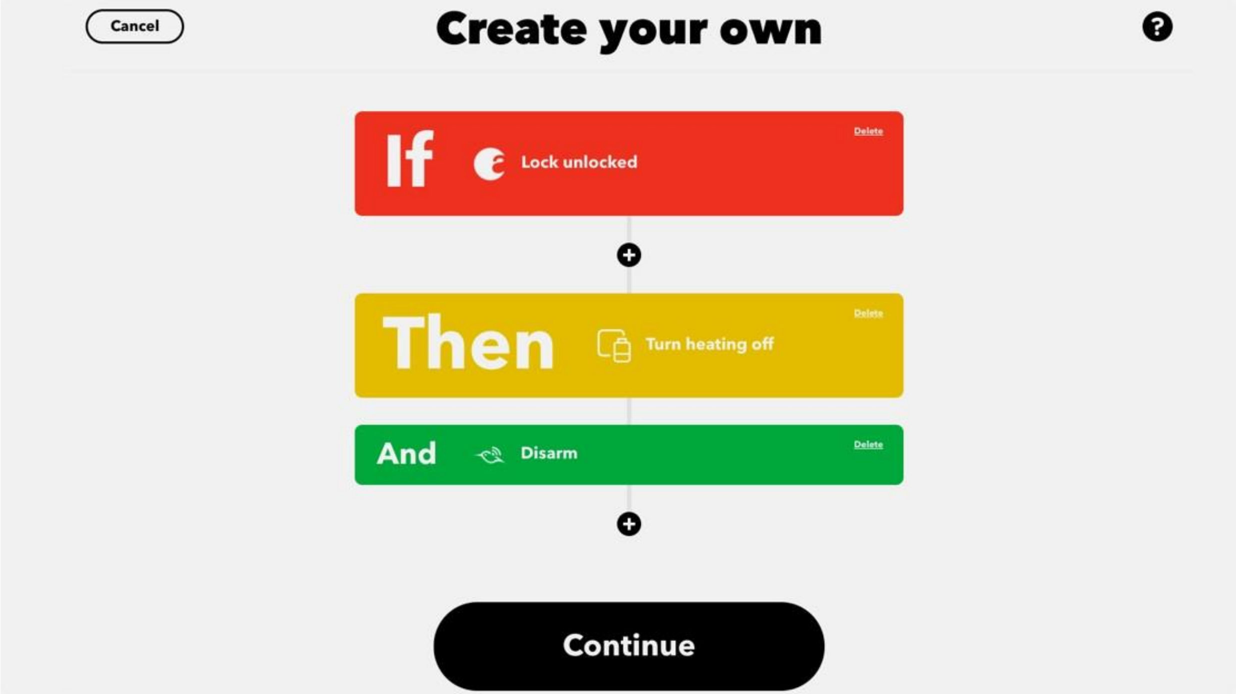
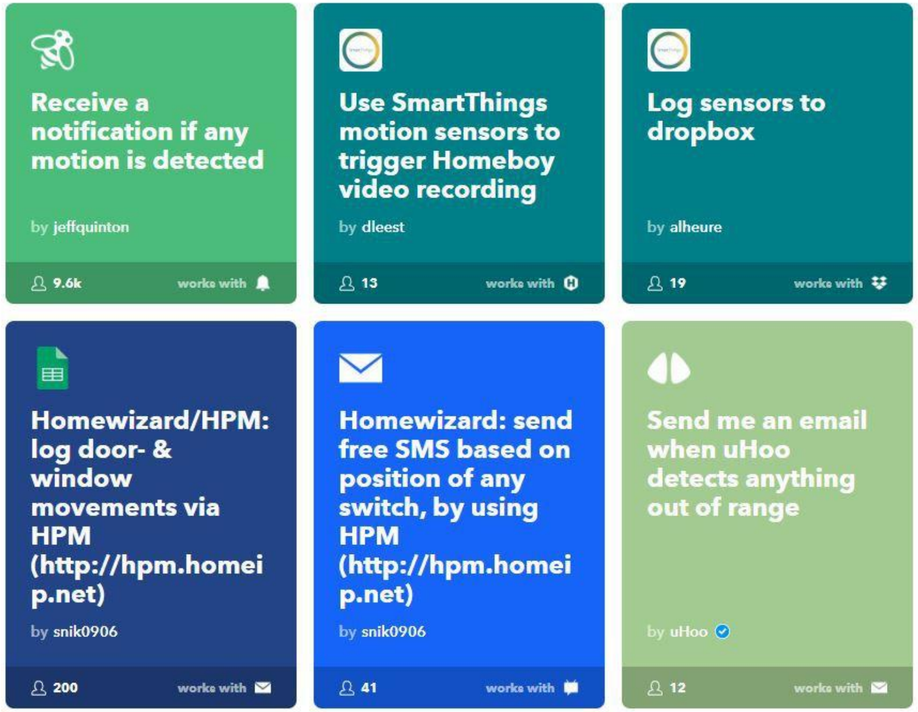
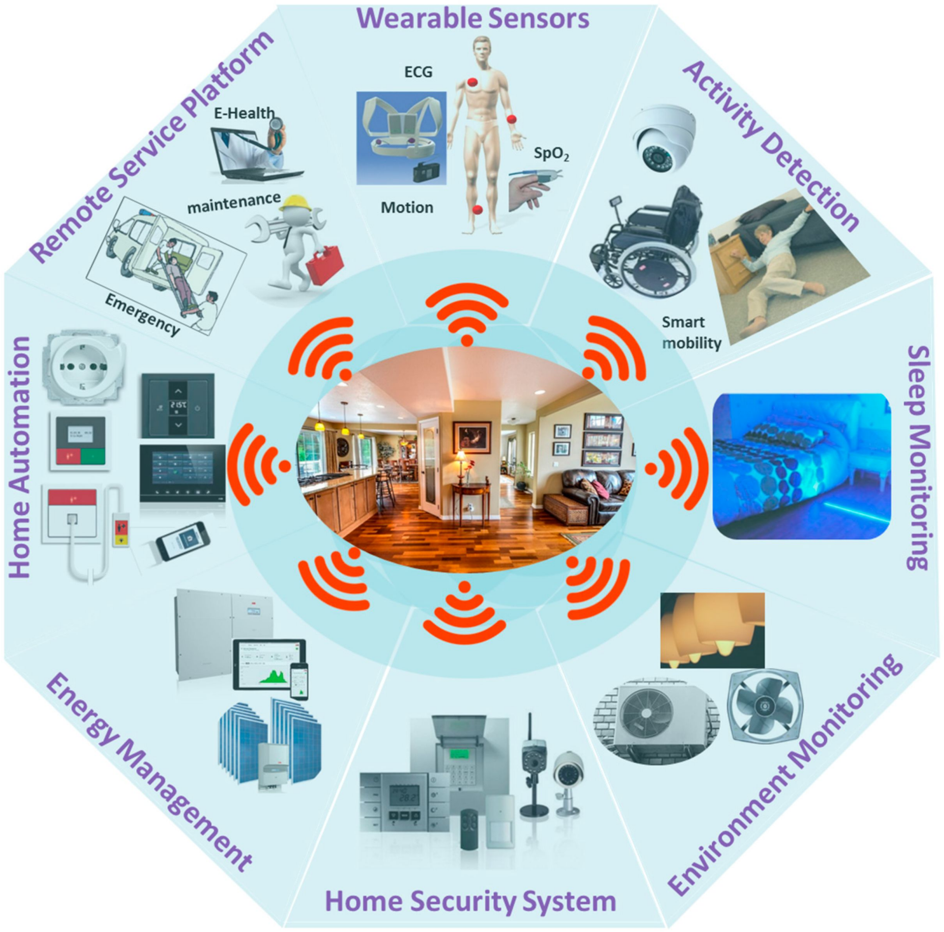
Automations





Domotics is more than automations

- IoT devices in smart-homes can provide way more applications than “automations”:
  - remote monitoring of subjects for healthcare applications
  - energy management
  - security
  - anomaly detection
  - …



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Context-Awareness

Understanding context

- Traditional software applications cannot understand the "context" of a request
- Users must make explicit all the request parameters
  - "What's the best route to go from X to Y, by car, considering that it's rush hour and that I hate highways?"
- Communicating with a computer is hard
- Limited interfaces (especially considering mobile devices)
- Ambient Intelligence systems aims at automatic mechanisms to acquire the context of the user!

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Ok but… what is “context”?

- Different interpretations (psychology, philosophy, computer science)

- From a vocabulary:
  - *“the set of circumstances or facts that surround a particular event or situation”*

- Many statements make sense only within their context
  - e.g., “How do I get there?”
  - the context should specify a) where the user is, b) where they want to go, c) how (on foot, on transportation)

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Early definitions

- Many definitions have been proposed for computer science applications:
  - By enumeration (proposals from different papers)
    - *location, time, surrounding people, season, temperature*…
    - *location, environment, identity and time*
    - *mood, level of attention, location, time, surrounding objects and people*
    - …
- However, such definitions are not general enough

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A common definition

- ***Anind Dey (2001) Understanding and Using Context. Personal and Ubiquitous Computing***:
  - *“Context is any information that can be used to characterize the situation of an entity. An entity is a person, place, or object that is considered relevant to the interaction between a user and an application, including the user and applications themselves.”*
- Simplifying: “All the data useful to adapt a service”
- General enough for our applications

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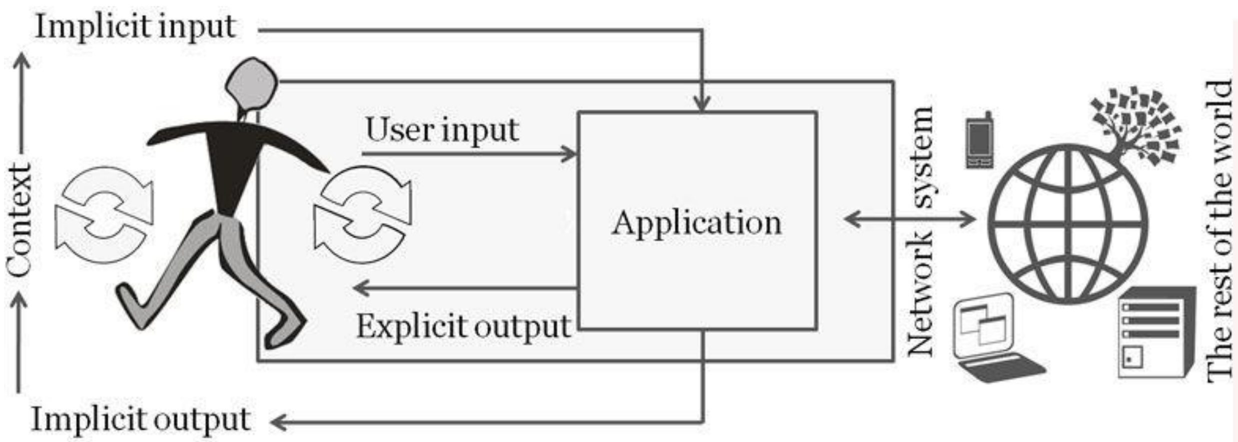
Context-Aware Systems

- A system is ***context-aware*** if it uses context to provide relevant information and/or services to the user
  - relevancy depends on the user's task!

- Ambient Intelligence systems are ***context-aware*** systems



Adaptiveness

- The property of a system to adapt to a given context for providing a better service/experience
- A **context-aware system** acquires context data to automatically adapt its behavior
- Very important in pervasive computing because of:
  - changes in user situation (location, time, activity, ...)
  - changes in environment (light, temperature, crowd, ...)
  - changes in network connectivity
  - changes in energy availability (battery)
  - etcetera

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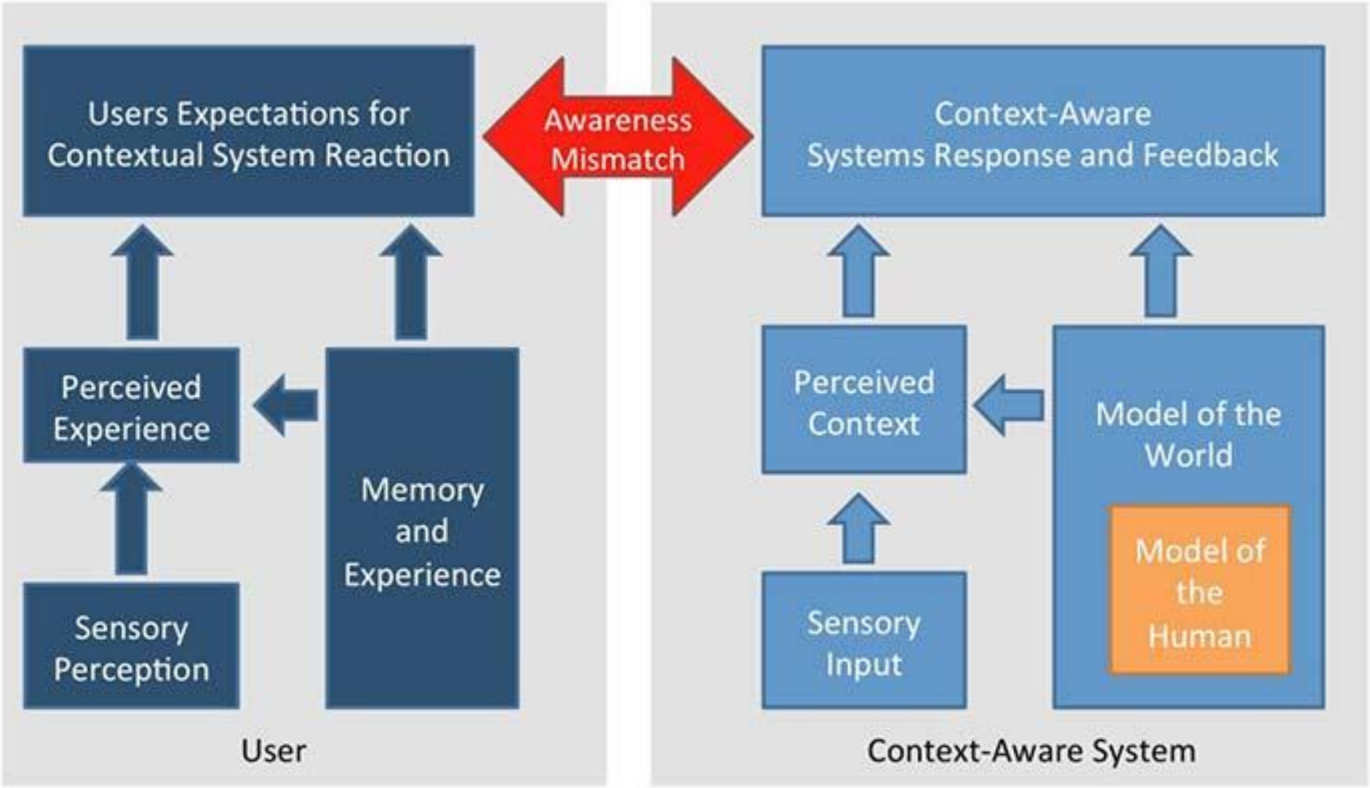
User perception of Context-Aware systems



- The quality of context-aware systems, as perceived by the user, is directly related to the awareness mismatch
    - i.e., mismatch between human perception and perceived context
- goal: minimizing awareness mismatch!

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Which contexts in Ambient Intelligence?

- Examples of contexts that may be useful in AmI systems:
  - human activities
  - human location
  - emotion
  - environmental conditions
  - energy consumption
  - connectivity
  - user preferences/habits
  - time
  - …

Classification of context

- A taxonomy

  (Not an exhaustive list)
  - U = user
  - S = sensors
  - D = device
  - O = other

Primary dimension	Secondary dimensions	Sources
User	Identity	U
	Physiological	S
	Emotional	S
	Interests	U, O
	Preferences	U, O
	Social	U, S, O
	Organizational	U, O
	Activity	U, S, O
Environment	Weather	S
	Temperature	S
	Humidity	S
	Light	S
	Noise	S
	Air pollution	S
	Surrounding persons / devices	S, O
Location	Physical	S
	Symbolic	S, O
Time	Absolute	D, S
	Granularity-based	D, S, O
Device	Capabilities	D
	Status	D
Connectivity		D, O

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Temporal context

- Not only
  - time of day, month, season, time zone, ...
- *Context history also* plays an important role
  - keeping track of past context data enables
    - deriving new context (e.g., by knowing a sequence of positions we can obtain a trajectory)
    - predicting context (e.g., observing a recurrent trajectory at a given time we can predict current destination)

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Obtaining context data

- **Low-level context**
  - Directly acquired from sensors or other sources
    - Example: Raw data from sensors, explicit user preferences from profiles, mobile device capabilities
  - Obtained by simple processing and/or fusion of raw data
    - Examples:
      - Taking bandwidth estimation by averaging over a set of samples in a time window
      - Deriving "Hot&Humid" context if the average of temperature values from different sources in the home are over 30°C and humidity is over 80%

Inferring context data

- **High-level context**
  - Derived by applying AI models:
    - Examples:
      - the activity "cooking a meal" inferred by smart-home sensors data
      - user's mood derived by combining data from the activities performed in the day, Galvanic Skin Response sensed at the wrist, and face recognition if available

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Applications of Ambient Intelligence Systems

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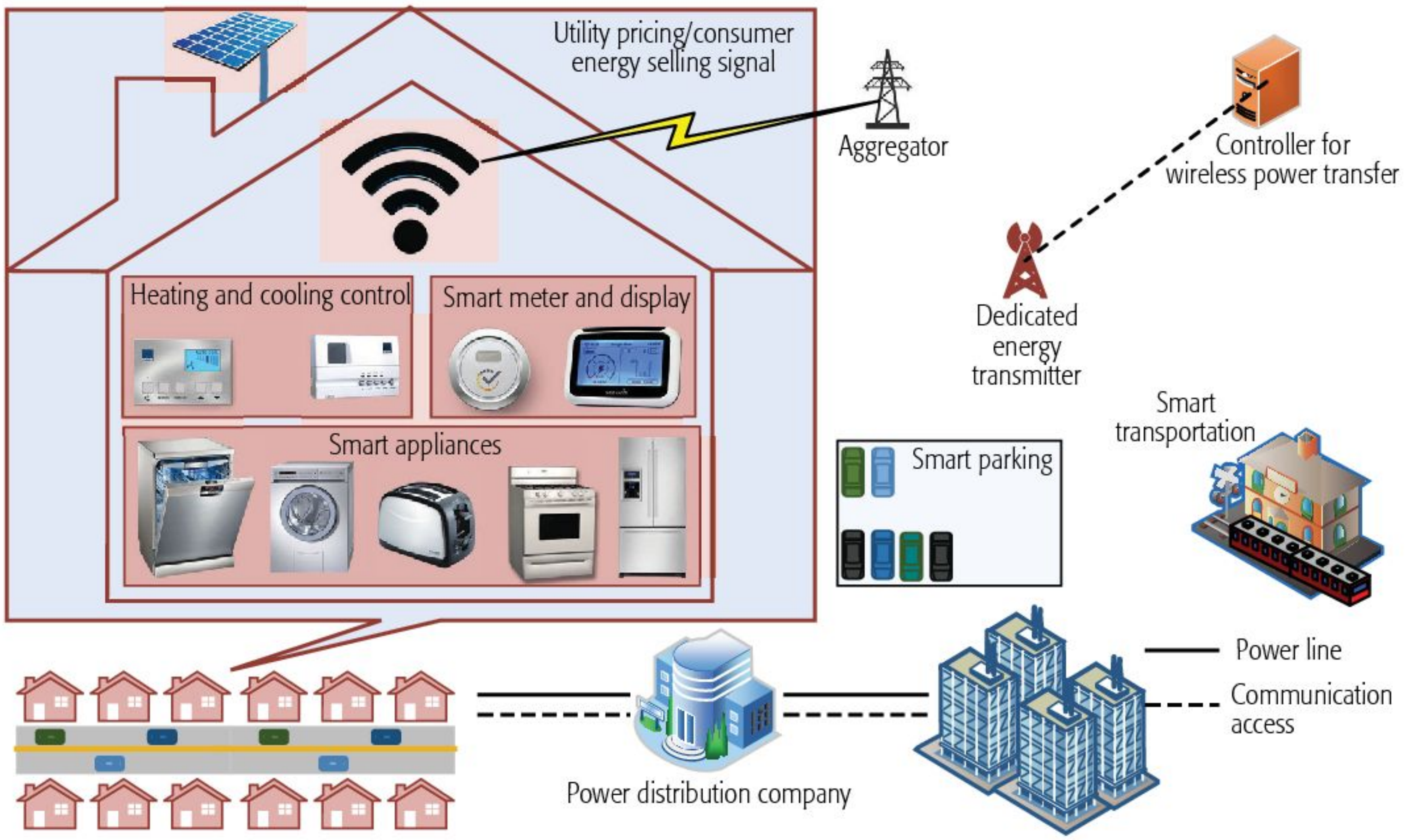
Domotics in Smart Homes

- Analyzing activities of people at home
  - to trigger automations
  - for ambient assisted living
  - energy management

Smart Energy Management

- Analyzing power consumption of home appliances
  - to suggest to the user energy consumption strategies
  - trigger home automations to save energy
  - optimize energy distribution for smart cities



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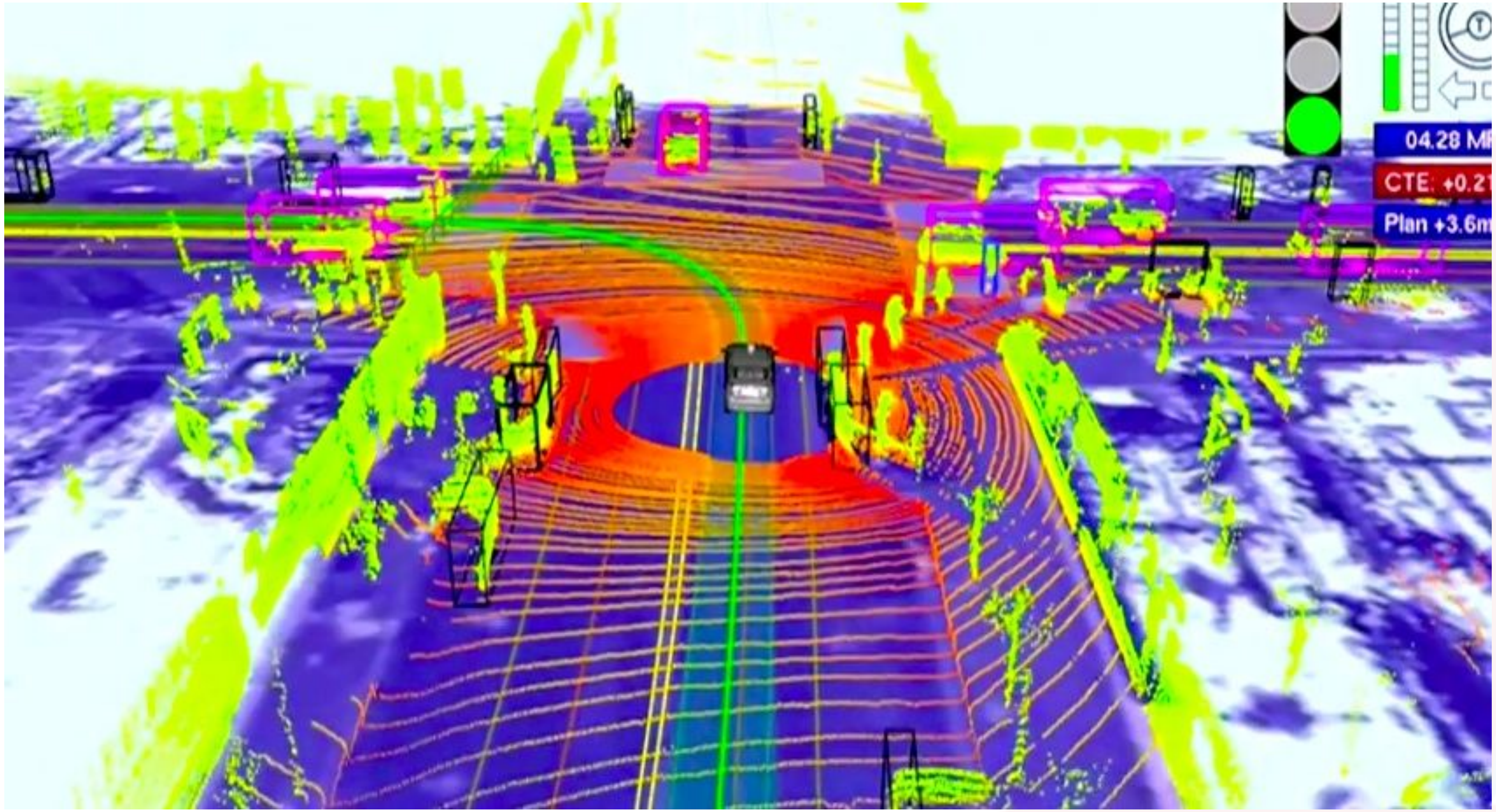
Smart Buildings

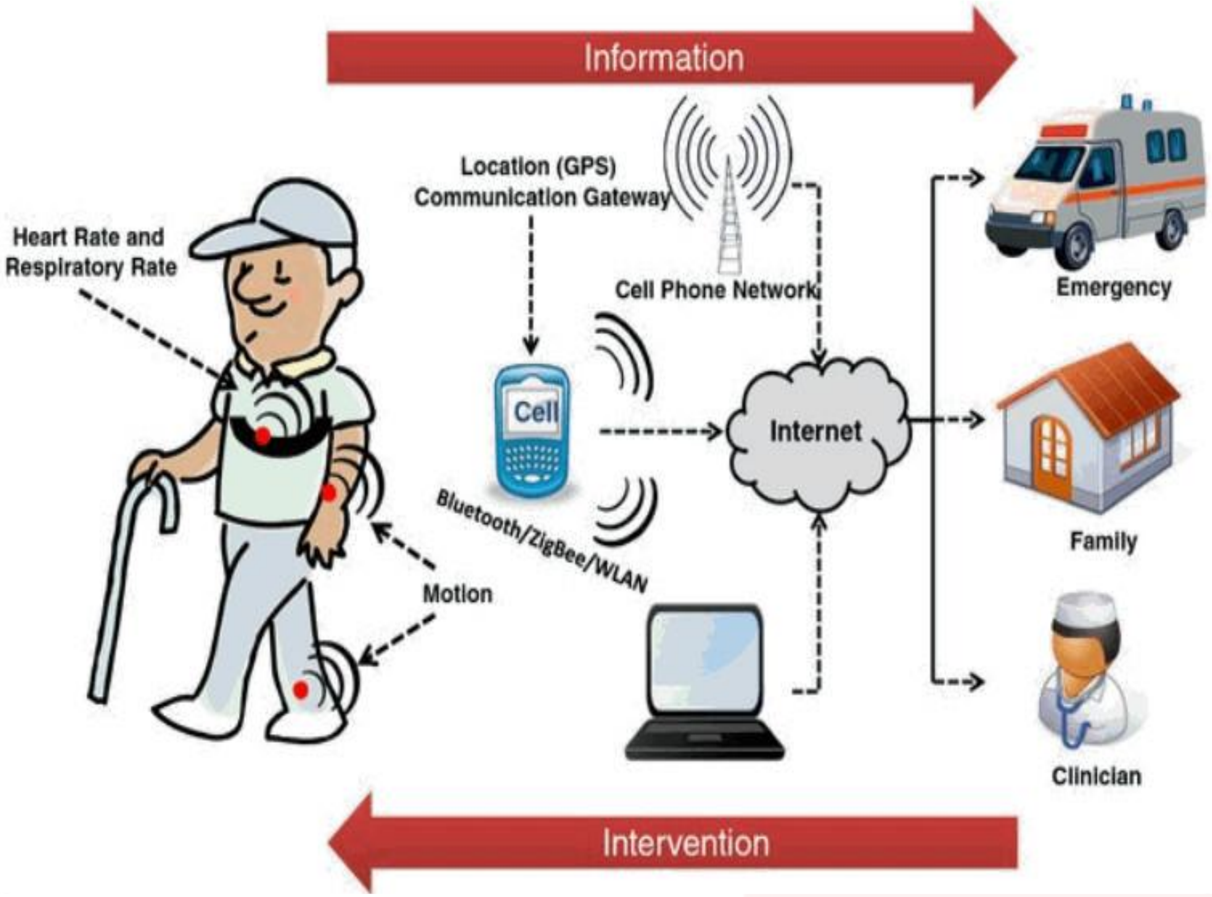
Pervasive Transportation

- Road sensors, car sensors, driver smartphones coupled with inter-vehicle communication enable new services
- **Goal**: self-driving vehicles, mobility optimization, emission reduction, hazard detection



Pervasive Health

- Wearable devices and environmental sensors enable continuous monitoring
- Goals:
  - early detection of health problems
  - monitoring health conditions in cronic diseases
  - facilitating rehabilitation at home
  - extended independent living (reminders, alarms, intervention, ...)



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Smart Cities

Some challenges in applying AI methods to Ambient Intelligence

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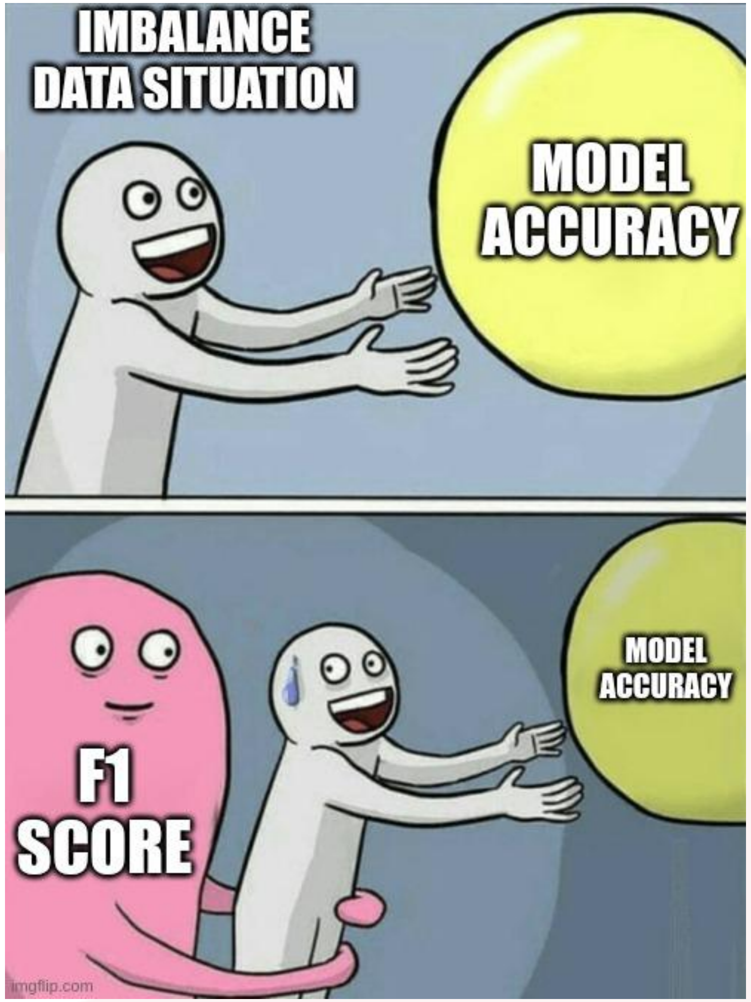
Unbalanced data

- Human Behavior is not usually associated with balanced data
  - for instance, considering human activities, some may be very rare (e.g., taking stairs) and others more common (e.g., walking)
- Hence, dealing with AmI data means having to do with unbalanced datasets (unless they were collected in a “controlled” but unrealistic way)



Labeled Data Scarcity

- Ambient Intelligence is about analyzing data collected from real users
- Obtaining large labeled datasets of human data is often prohibitive
  - annotation is costly, time-consuming and intrusive!
- Several approaches have been proposed to mitigate this problem, we will see some of them

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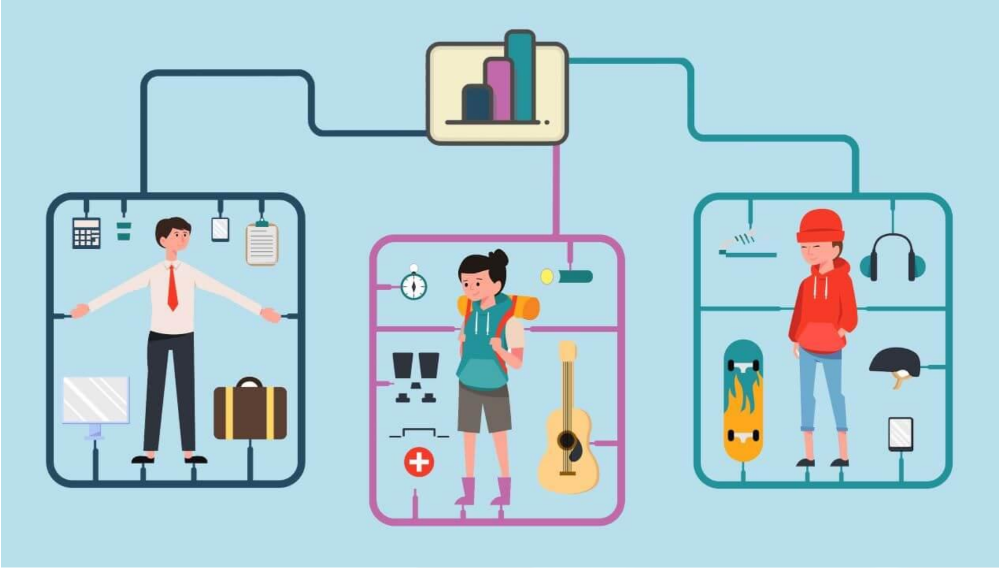
Personalization

- Ambient Intelligence requires AI models that should be tailored to the target subject/environment
  - Differently from other domains (e.g., computer vision) where pre-trained models may often be used even without fine-tuning!
- This is due to the fact that sensor data patterns significantly vary for different subjects/environments
- For instance:
  - each smart home or building may have its peculiarities
  - each subject may have peculiar way of performing activities or daily habits
- Personalizing the AI model is often a crucial aspect!
  - (even if this is challenging due to the labeled data scarcity problem)



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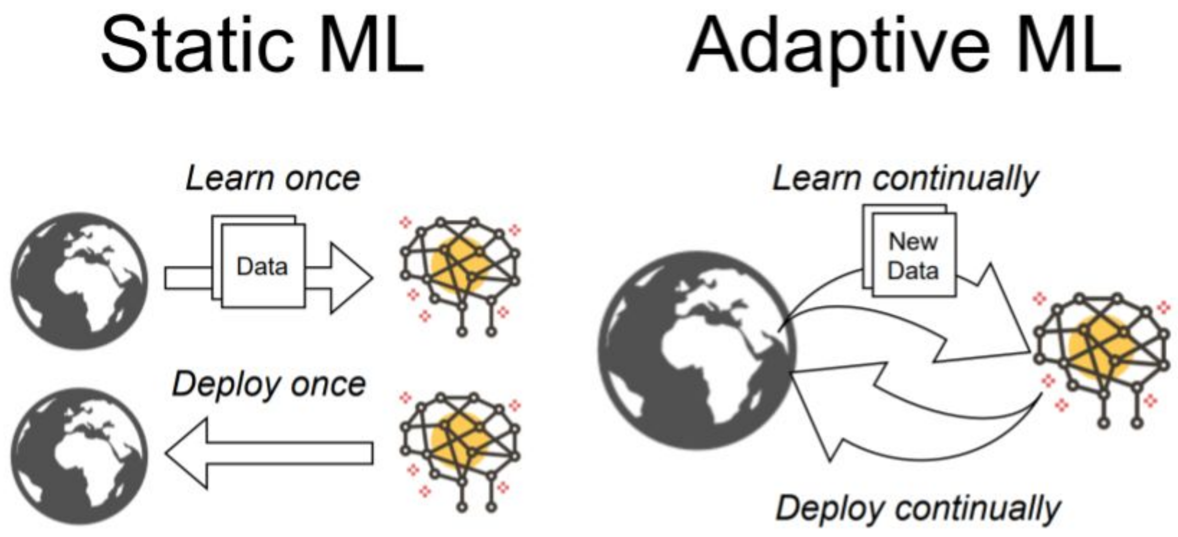
Continual Learning

- Ambient Intelligence systems generate high amounts of sensor data
- Patterns and user behaviors may change with time
    - a model trained once only at the begin may not enough!
- There is the need of continuously update the model without retraining from scratch
    - problem: neural networks tend to ***forget*** old information



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Privacy

- Data collected in ambient intelligence scenarios may be sensitive, for instance:
  - users' activities and habits
  - energy consumption patterns
  - health disease
- These information may be disclosed to untrusted third parties (e.g., cloud services, malicious attackers)
- It is hence important to have privacy in mind when designing Ambient Intelligence solutions!

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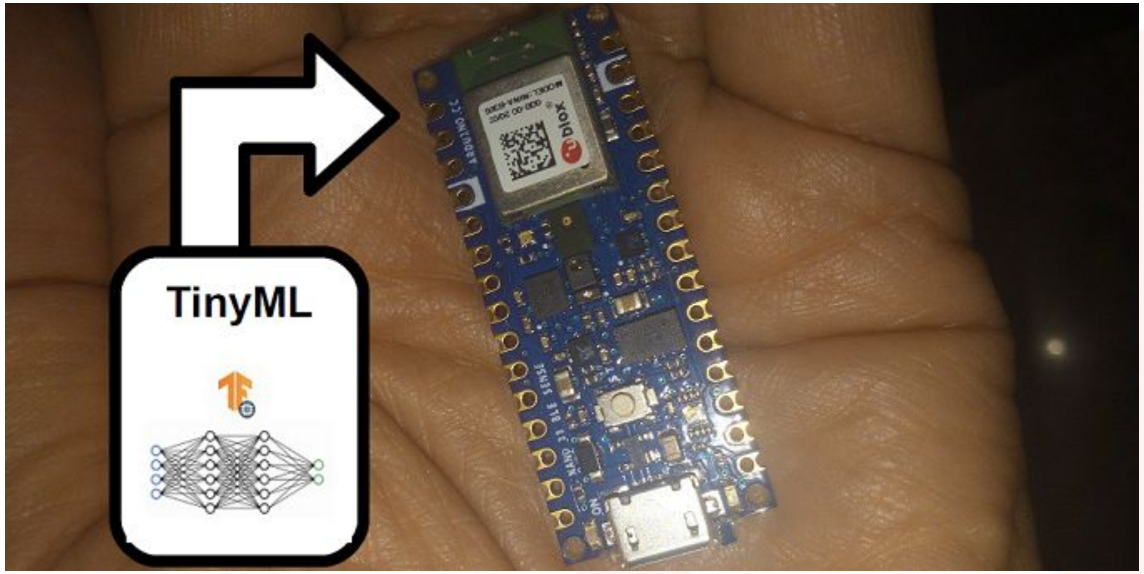
Resource-constrained devices

- Sometimes it may be necessary to run deep learning models on resource-constrained devices instead of using cloud services (e.g., smartphones or micro-controllers)
  - to reduce latency
  - to work offline
  - to guarantee privacy
  - to reduce costs
- The challenge is to reduce the complexity of the models so that they can use limited resources and consume limited energy



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What will we see in this course?

In this course

- We will mainly investigate AI methods to analyze low-level ***sensor data time series***
- **Objective:** infer high-level ***context*** that is crucial for AmI scenarios
  - (e.g., human behavior, energy consumption, indoor positioning)
- Smart-home data and mobile/wearable settings
- We will also investigate advanced topics in this area, such as:
  - Self-Supervised Learning
  - eXplainable AI
  - Federated Learning
  - Graph Neural Networks (GNNs)
  - Generative AI

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References

[1] *Cook, D. J., Augusto, J. C., & Jakkula, V. R. (2009). Ambient intelligence: Technologies, applications, and opportunities. Pervasive and mobile computing, 5(4), 277-298.*

[2] https://www.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/context-aware-computing-context-awareness-context-aware-user-interfaces-and-implicit-interaction

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